7. Verify whether the following compound propositions are tautologies or contradictions or contingency

   (i) $(p \wedge q) \wedge \neg(p \vee q)$ (ii) $\big((p \vee q) \wedge \neg p\big) \rightarrow q$

   (iii) $(p \rightarrow q) \leftrightarrow (\neg p \rightarrow q)$ (iv) $\big((p \rightarrow q) \wedge (q \rightarrow r)\big) \rightarrow (p \rightarrow r)$

8. Show that (i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$ (ii) $\neg(p \rightarrow q) \equiv p \wedge \neg q$.

9. Prove that $q \rightarrow p \equiv \neg p \rightarrow \neg q$

10. Show that $p \rightarrow q$ and $q \rightarrow p$ are not equivalent

11. Show that $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

12. Check whether the statement $p \rightarrow (q \rightarrow p)$ is a tautology or a contradiction without using the truth table.

13. Using truth table check whether the statements $\neg(p \vee q) \vee (\neg p \wedge q)$ and $\neg p$ are logically equivalent.

14. Prove $p \rightarrow \big(q \rightarrow r\big) \equiv (p \wedge q) \rightarrow r$ without using truth table.

15. Prove that $p \rightarrow (\neg q \vee r) \equiv \neg p \vee (\neg q \vee r)$ using truth table.

AXZN2

EXERCISE 12.3

Choose the correct or the most suitable answer from the given four alternatives.

1. A binary operation on a set S is a function from

   (1) $S \rightarrow S$ (2) $\big(S \times S\big) \rightarrow S$ (3) $S \rightarrow \big(S \times S\big)$ (4) $\big(S \times S\big) \rightarrow \big(S \times S\big)$

2. Subtraction is not a binary operation in

   (1) $\mathbb{R}$ (2) $\mathbb{Z}$ (3) $\mathbb{N}$ (4) $\mathbb{Q}$

3. Which one of the following is a binary operation on $\mathbb{N}$?

   (1) Subtraction (2) Multiplication (3) Division (4) All the above

4. In the set $\mathbb{R}$ of real numbers '$*$' is defined as follows. Which one of the following is not a binary operation on $\mathbb{R}$?

   (1) $a * b = \min(a \cdot b)$ (2) $a * b = \max(a, b)$

   (3) $a * b = a$ (4) $a * b = a^b$

5. The operation $*$ defined by $a * b = \dfrac{ab}{7}$ is not a binary operation on

   (1) $\mathbb{Q}^+$ (2) $\mathbb{Z}$ (3) $\mathbb{R}$ (4) $\mathbb{C}$

6. In the set $\mathbb{Q}$ define $a \odot b = a + b + ab$. For what value of y, $3 \odot \big(y \odot 5\big) = 7$?

   (1) $y = \dfrac{2}{3}$ (2) $y = \dfrac{-2}{3}$ (3) $y = \dfrac{-3}{2}$ (4) $y = 4$

7. If $a * b = \sqrt{a^2 + b^2}$ on the real numbers then $*$ is

   (1) commutative but not associative (2) associative but not commutative

   (3) both commutative and associative (4) neither commutative nor associative

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